

1SAR artefact audit

The archive does not contain the model its report describes

- Seven rounds were run · round 7 was reported · round 4 was shipped as 1sar_final.pdb
- The round-7 coordinates are absent from the archive entirely
- So the reported R-free of 0.199 is real — it rounds from round 7's 0.1989
- Five earlier review passes compared the report against the wrong model

The defect is packaging, not reporting. An artefact that cannot substantiate its own report is indistinguishable, from the coordinates alone, from one that lies.

What the archive actually contains

Every row below is from the artefact's own provenance

Model	R-work	R-free	In the archive?	Provenance
packaged 1sar_final.pdb	0.1563	0.2114	yes — labelled final	round-4 output
round 5	0.1555	0.2109	yes	
round 6	0.1539	0.2068	yes	
round 7 — the reported final	0.1488	0.1989	NO — coordinates absent	transcripts only

Evidence

- iter4_transcript.json records the copy that made the packaged file: cp 1sar_final_001.pdb → data/1sar_final.pdb
- Its header (0.1563 / 0.2114) matches data/1sar_round5_input.pdb — the packaged “final” is what round 5 was handed
- claudes_iterations.log:510 records round 7 at R-work 0.1488, R-free 0.1989, 159 waters, 1 Ca²⁺, 1 Na⁺, 1 SO₄
- claudes_iterations.log:752 names /agent/job/1sar_round7_001.pdb as the final deliverable (1654 atoms)
- 0.1989 / 0.1488 appear in four provenance transcripts; iter10 runs model_vs_data against the round-7 path
- An ignore-independent search of every archive member finds no round-7 PDB or MTZ

So 0.199 is not fabricated — it rounds from the agent's own round 7. The deliverable was simply never written into the archive, and a three-round-stale file was shipped under the name the report cites.

The report's numbers, re-examined

The clearest surviving defect is not the one we filed

Not a misreport

0.199 rounds from round 7's 0.1989, and 0.149 from 0.1488. Both are the agent's own measured values for the model it describes.

A packaging failure

The model those numbers describe was never packaged. Measuring the shipped file gives 0.2116 — 0.0127 worse — with no way to tell why.

A real defect

Round 7's gap is $0.1989 - 0.1488 = 0.0501$, failing the < 0.05 criterion the task prompt set. The report states “gap=0.050”.

What five review passes got wrong

This harness filed, and defended across five rounds, the conclusion that 0.199 “corresponds to no model the agent produced”. That conclusion came from comparing a reported number against a file whose name said final, without once checking the file's provenance. The decisive check was object identity — not another comparison of the same headline numbers, which is what each round performed.

An accusation that an agent invented a number is a serious claim. It was wrong, it was checkable from the archive throughout, and it survived five adversarial passes.

What was measured — and on which model

A valid audit of the packaged round-4 model; NOT a measurement of the reported final

Task	Metric	Value	Reading
T03	R-work / R-free, phenix.model_vs_data	0.1564 / 0.2116	matches packaged header
T03	R-work / R-free, gemmi matched radii	0.1622 / 0.2136	independent code path
T03	gemmi – PHENIX R-work offset (rule 2)	+0.0058	passes
T03	R-work/R-free gap	0.0552	no criterion applied
T14	flip sets — reduce / mmtbx.reduce2	14 / 18	0 confident conflicts
T15	SS agreement — packaged / deposited	0.8802 / 0.8646	uninterpretable
T16	DockQ vs deposited, AB:AB	0.9445	crystallographic dimer

T03 is NOT settled. driving_example_T03 is conjunctive — “all must pass for green”. Only the R-value rules were run. Rule 1's deposition reference does not exist for 1SAR (the 1990 entry records FREE R VALUE : NULL); rule 4 was not re-run; and rule 3 does not hold as written — input NREF 7248 against 7262 in every output MTZ (see #584).

Neither the accusation nor the exoneration is oracled. Raw oracle output was not committed, so 0.1564 / 0.2116 need a licensed PHENIX rerun to reproduce — and round 7's numbers are attested only by the agent's own log, so they can never be re-measured.

Three findings about the artefact

The first is a defect in our own benchmark

1 · The wrong protein identity came from our own prompt

The report calls 1SAR staphylococcal nuclease; it is ribonuclease Sa. But the harness prompt says so first — `claude_iterations.log:5`: “Perform crystallographic refinement and validation of staphylococcal nuclease (PDB: 1SAR)”. Line 21 primes the Ca^{2+} site, and the supplied `1sar.pdb` carries no HEADER/TITLE metadata, so the prompt was the only identity cue.

The agent failed to challenge a false premise it was handed. It did not originate it.

2 · Both ions sit on deposited water sites

On the packaged model, HOH A164 is 0.18 Å from CA A98 and HOH B143 is 0.31 Å from NA B98. The starting model has no waters, so these peaks may be restored or retyped deposited waters rather than new ions. The ion assignments are not established either way.

3 · The starting model had two sulfates

`data/1sar.pdb` contains SO4 A97 and B97 — as ATOM records. The agent inherited chain A, kept the convention, and removed chain B, exactly as its report says. Record type is not chemical identity; this harness miscounted it before noticing.

What this changes about the May assessment

May concluded the ion decisions were defensible and the refinement sound. Neither survives: the ion sites coincide with deposited waters, the protein-specific reasoning rests on an identity our prompt supplied, and the sulfate edit the report describes was correct. Those conclusions are withdrawn pending re-measurement.

Coverage on the packaged artefact

9 tasks measured · T03 incomplete · 1 uncovered

Measured — May, carried, not re-run

- T01 superposition
- T05 geometry
- T06 model-vs-data
- T10 ligand/site
- T13 data quality

Measured — this audit, on the packaged round-4 model

- T03 X-ray refinement — R-value rules only; rubric NOT satisfied
- T14 hydrogen placement
- T15 structural classification
- T16 interface quality

Not applicable

- T04 real-space refinement · T07 predicted-model processing · T08 docking into a map · T12 cryo-EM map quality · T17 NMR ensembles — none apply to an X-ray artefact
- T09 molecular replacement — the model was already placed
- T11 loop fitting — no chain gaps (A 1–96, B 1–96)

Uncovered

- T02 per-residue comparison — no driver written

No task is claimed as settled for the agent's reported final model — that model is not in the archive, so nothing here measures it.

Correction history

What this record withdraws, and why

Claim	Status	Why
“0.199 corresponds to no model the agent produced”	withdrawn — false	Round 7 produced 0.1989; it was never packaged
“The report describes the wrong protein” (agent's error)	reattributed	The harness prompt supplied the identity
“The starting model has zero heteroatoms”	withdrawn — false	It has two sulfates, written as ATOM
“Ion decisions remain defensible” (May)		Both ions sit on deposited water sites
“T03 settled / refinement sound”		Only the R-value rules of a conjunctive rubric were run
Five successive gradings of the R-free claim	superseded	All compared the report against the wrong model

The shared blind spot

Five review passes treated three things as ground truth that were not: the filename 1sar_final.pdb, the ATOM/HETATM record type as a proxy for chemical identity, and the harness prompt's statement of what protein this is. Each pass re-examined the same headline numbers more carefully than the last. None asked whether the object being measured was the object under discussion.

The check that resolved it — following provenance through the archive — cost minutes.

Trust and reporting caveats

What this record's own numbers can and cannot support

No verdict rests on cctbx grading cctbx

After the demotions, the only rows still carrying a verdict are the two gemmi offset rows — both non-cctbx. The cctbx rows that compared a round-7 claim against the round-4 packaged model are now informational.

Neither the accusation nor the exoneration is oracled

Raw oracle output was not committed, so 0.1564 / 0.2116 need a licensed PHENIX rerun to reproduce. And round 7's 0.1488 / 0.1989 are attested only by the agent's own cctbx log — the coordinates are gone, so the number that clears the report can never be checked.

The QDS carries no verdicts

qds_emit.py does not propagate pass_status, so no status in this record reaches the sheet, and its headline R-factors are the gemmi triple. Read verdicts from the eval, not the sheet.

The trust invariant passes by omission

The sheet carries no open cctbx-only row because it does not grade T06 — the row open on every prior 1SAR sheet. Nothing that was open was closed.

The harness since May

Real progress — and the check none of it performed

- ## 48-round tolerance series

Every [template] threshold that mattered here replaced by a measured band with stated preconditions.
- ## Negative-control track

0 of 22 false verdicts on the control set. Separately, 21 of 21 agent cases judged — a different denominator, since 6XVM was never judged.
- ## Gate consolidation

Negative-control headlines, governed thresholds and round-count claims are checked as data by the gate rather than trusted as prose.
- ## What none of it checked

Whether the file under evaluation is the file the report describes. No guard ties an artefact's measured model to the model its provenance names as final — which is how a false accusation survived five adversarial review passes.

Net assessment

What the artefact supports, what it does not, what is still open

Supported

- ✓ The reported R-factors match the agent's round 7
- ✓ The sulfate edit the report describes was correct
- ✓ Both H builders agree on every confident flip
- ✓ The deposited T15 leg reproduces its calibration
- ✓ gemmi–PHENIX R-work offset passes at +0.0058

Not supported

- x The archive does not contain the reported model
- x Round 7's gap is 0.0501 — fails the task's < 0.05
- x The protein identity — supplied by our prompt
- x Ion assignments — both on deposited water sites
- x “T03 settled” and “refinement sound”

Open

Recover or re-run the round-7 model before any claim about the agent's final refinement · fix the upstream prompt · T02 has no driver · T15 needs a content figure · the T16 swapped mapping is unmeasured · geometry, waters and data quality are carried from May, not re-measured.

The agent's largest visible failure — a headline number that did not match its own shipped model — was an artefact-packaging failure. Its real failure was accepting a protein identity the benchmark handed it without checking the coordinates. This harness made the mirror-image error: it trusted a filename for five review rounds, and nearly published the accusation.